

Rohit Punjabi

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Professional Summary

Software Engineer with experience delivering production-ready features across multiple releases. Skilled in C++, TypeScript, and modern frameworks, with a strong background in Unreal Engine and Unity. Adept at building scalable backend and client systems, collaborating across disciplines, and applying data-driven approaches to create efficient, player-focused solutions.

Work Experience

Software Engineer | Shrapnel

Feb 2024 to Sept 2024 | Seattle WA USA

- Developed a telemetry system in Unreal 5.3 (C++) with a MongoDB backend to track player metrics (kills, weapon usage, movement, power-ups), enabling data-driven gameplay improvements.
- Designed and implemented a social system in Unreal 5.3 (C++ and Blueprints) supporting unlimited friend requests, boosting player engagement, and team play.
- Built a weapon skin customization system for four gun types with regularly updated skins, enhancing personalization and in-game variety.
- Led development of an internal analytics dashboard using TypeScript and React, reducing bug detection and analysis time by ~70% and improving player behavior insights.
- Contributed to seven early access releases, collaborating with 5+ cross-disciplinary teams to deliver features across C++, TypeScript, and Unreal 5.3.

Tools Software Engineer | Shrapnel

Aug 2023 to Feb 2024 | Seattle WA USA

- Improved productivity by 80% and prevented data loss by developing key features.
- Led a UI/UX refresh to align tool design with game UI, adding new views for managing projects and improving usability.
- Enhanced stability by fixing a critical bug that reduced crashes by 30%.
- Built a web-based moderation tool in TypeScript/React, streamlining content review into a 2-step process, improving efficiency for senior users.
- Partnered with designers, artists, and power users to gather feedback and deliver high-impact improvements.
- Contributed to client-side tool development in a 3-person team using C++, C#, Unity3D, GitHub, and TypeScript, enabling backend developers to focus on server-side work.

Education

Master of Science in Computer Science

DigiPen Institute of Technology | Redmond, WA, USA | May 2023

Bachelor of Engineering in Information Technology

Thadomal Shahani Engineering College | Mumbai, MH, India | May 2021

Skills

Programming Languages: C++, C#, TypeScript, React, Python, SQL

Tools: GitHub, Perforce, Jira, Docker, Postman

Game Engines: Unity3D, Unreal 5.3

Projects

3D Custom Game Engine

Sept 2022 to Apr 2023

- Built a 3D game engine from scratch in an Agile development environment.
- Implemented a stack-based game state manager for seamless transitions.
- Designed a car steering mechanism with custom algorithms for angle calculation.
- Created an Audio System using FMOD Core API, enabling advanced audio functionality.
- Designed all UI textures to align with the game's theme and improve player experience.

Convolutional Neural Network (CNN)

Feb 2023

- Developed a CNN to classify images for image processing using Python, TensorFlow, and NumPy.
- Pre-processed data using normalization and image resizing techniques to optimize the model's performance.
- Visualized the results using accuracy and validation curves to gain insights into the model's performance.

2D Component Based Game Engine

Jan 2022 to Apr 2022

- Built a robust 2D game engine from scratch with modular architecture.
- Implemented collision detection algorithms (AABB, circle collision).
- Developed a custom behavior tree system with reusable nodes (conditionals, composites, decorators).
- Created an input handling system using Lua scripting for flexible mapping of actions.